

Abhinav Kumar

AI QA Automation Engineer | SDET | Test Automation & Gen AI/LLM Integration

New Delhi, India | abhinavabhi0805@gmail.com | +1-628-250-5140 / +91-9015779490 |
linkedin.com/in/abhinav-kumar-248263112

Professional Summary

QA Automation Engineer / SDET with 5 years of experience designing, developing, and maintaining test automation frameworks for complex enterprise systems in agile environments. Builds end-to-end automation with Selenium, Playwright, Java, and TypeScript using the Page Object Model, integrated with CI/CD pipelines to reduce regression testing time by 70% and sustain 95% automation coverage. Collaborates with developers, analysts, and Quality Engineer teammates on test coverage, regression strategies, and code reviews, and mentors teammates to grow their automation and technical skills. Experienced across HRMS, compliance, and fintech/investment domains with a remote-first working style suited to distributed U.S. teams.

Technical Skills

- **Test Automation Frameworks:** Selenium WebDriver, Playwright, Playwright MCP, TestNG, Page Object Model (POM)
 - **Programming Languages:** Core Java, TypeScript
 - **CI/CD & DevOps:** GitHub Actions, CI/CD Pipeline Integration, Automated Nightly Regression, Allure Reporting
 - **API & Performance Testing:** REST API Testing (Postman), JMeter Performance & Load Testing
 - **Databases & SQL:** MySQL Database Testing
 - **Web Technologies:** SharePoint Online (Microsoft 365), REST APIs
 - **Methodologies:** Agile/Scrum, Sprint Planning, Code Reviews, Requirement Traceability Matrix (RTM)
 - **AI / Gen AI & LLM (workflow enabler):** AI-Assisted Test Generation, Claude Code, ChatGPT, GitHub Copilot, RAG Pipeline Development, n8n Workflow Automation, LLM Testing & Validation (Llama 2)
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Professional Experience

SDET / AI QA Automation Engineer | Sphinx Worldbiz Ltd. — October 2021 – Present

HMEL ISO Compliance & STAR Portals (SharePoint Online) | Feb 2026 – Present | Playwright · TypeScript · GitHub Actions · Allure · RAG · n8n · Gen AI

- Built a RAG pipeline to query compliance documentation, improving test case traceability and cutting manual lookup time.
- Designed n8n workflows to trigger nightly regression runs and route Allure reports/pass-fail alerts to JIRA and Slack.
- Leveraged Claude and ChatGPT for test case generation, documentation, and locator optimization; led UAT sessions validating migrated data and integrations.
- Built Playwright/TypeScript end-to-end automation for service-request workflows, including multi-level approvals, using POM to produce 150+ reusable test scripts.
- Integrated the test suite into GitHub Actions CI/CD for automated nightly regression, cutting manual testing effort by 70%.

HMEL In-House Portal Migration & Testing (Client Site) | Mar 2025 – Jan 2026 | Playwright · TypeScript · GitHub Actions · Allure · n8n · Gen AI/LLM

- Used n8n to automate migration validation across 60+ portals and notify stakeholders of sign-off status via JIRA and email.
- Maintained 95% automation coverage using LLM-powered flaky-test analysis, and created test scenarios and RTMs with Generative AI.
- Built and optimized Playwright/TypeScript automation suites for 60+ migrated SharePoint portals, using ChatGPT and Claude to accelerate script generation.
- Designed scalable, AI-assisted automation frameworks enabling parallel execution and maintainable architecture.

Nature Alpha — Biodiversity Investment Platform | Dec 2024 – Mar 2025 | JMeter · REST API · JIRA · GitHub · Gen AI (ChatGPT)

- Executed performance and load testing with JMeter to assess system behavior under varied traffic conditions.
- Developed structured test plans and cases aligned with business requirements, using ChatGPT to accelerate test-plan drafting.
- Documented and tracked defects in JIRA through full resolution lifecycle, collaborating with cross-functional Agile teams.

Contractor Safety Management Portal | Oct 2024 – Dec 2024 | Selenium · Java · TestNG · Maven · JIRA · REST API · GitHub

- Implemented a Maven/TestNG automation framework for continuous testing of safety compliance modules.
- Designed and executed automated tests for contractor onboarding, work permit issuance, and safety violation handling.
- Performed end-to-end functional testing and managed the defect lifecycle in JIRA.

Dream Collector — AI-Powered Goal Achievement Platform | Feb 2024 – Sep 2024 | LLM Integration Testing · Mobile Testing · JIRA · Gen AI (Copilot)

- Developed automated test cases for AI-driven features powered by Llama 2 LLM integration, validating AI-generated content and dream-creation workflows.
- Executed functional, integration, and regression testing for OTP authentication and user profile management.
- Validated user stories in JIRA against acceptance criteria defined in BRD and FRD documents.

SHRM Pro — Human Resource Management System (HRMS) | Oct 2021 – Feb 2024 | Selenium · Java · REST API · Postman · Maven · JIRA

- Developed automated UI tests with Selenium/Java and POM, aided by AI code generation.
- Built a REST API testing framework in Postman covering authentication, authorization, CRUD operations, and data validation.
- Automated end-to-end workflows for employee data, payroll, and benefits administration modules.
- Used Langflow to prototype and validate low-code AI agent workflows for HR query automation before development handoff.

Key Achievements

- Built reusable automation frameworks from scratch, improving overall team productivity by 50%.
- Migrated and validated 60+ SharePoint portals with zero production defects.
- Sustained 95% automation coverage across enterprise applications through AI-assisted test maintenance.
- Pioneered adoption of Generative AI and workflow-automation tools (Claude Code, Copilot, RAG, n8n, Langflow) in QA workflows, cutting test-documentation effort by 40%.

Education

Bachelor of Engineering (B.E.) — Rajiv Gandhi Technical University (RGTU), Bhopal, M.P., India